Day 18 Of Communication System Class

Topic Covered

1. Hilbert Transformation [PYQ]
2. Pre-emphasis & De-emphasis [PYQ]

Where these topics lie in recent syllabus

2.2 AM for single & double tone message, carrier & sideband components, power in carrier & sideband components, bandwidth & power efficiency, Hilbert transform

3.6 Stereo FM, spectral details, pre-emphasis and de-emphasis network

—

Hilbert Transformation [PYQ]

Hilbert transform of a signal x(t) is defined as the transform in which phase angle of all components of the signal is shifted by $\pm 90^{\circ}$.

Hilbert transform of x(t) is represented with $\hat{x}(t)$,and it is given by

$$\hat{x}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(k)}{t-k} dk$$

The inverse Hilbert transform is given by

$$\hat{x}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(k)}{t-k} dk$$

x(t), $\hat{x}(t)$ is called a Hilbert transform pair.

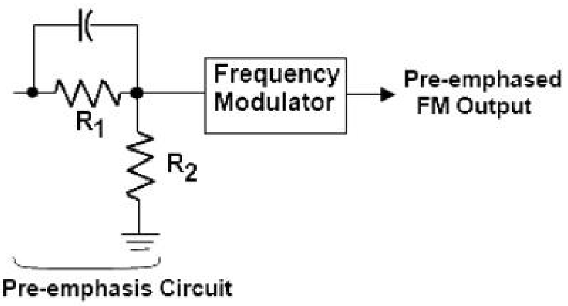
Properties of the Hilbert Transform

A signal x(t) and its Hilbert transform $\hat{x}(t)$ have

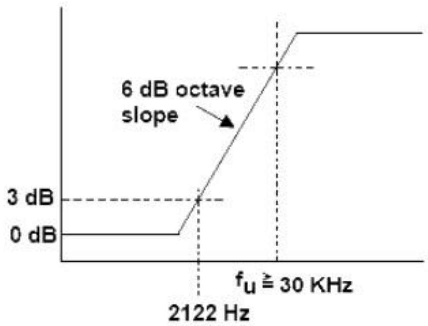
- The same amplitude spectrum.
- The same autocorrelation function.
- The energy spectral density is same for both x(t) and $\hat{x}(t)$.
- x(t) and $\hat{x}(t)$ are orthogonal.
- The Hilbert transform of $\hat{x}(t)$ is -x(t)
- If Fourier transform exist then Hilbert transform also exists for energy and power signals.

Pre-emphasis & De-emphasis [PYQ]

Pre-emphasis is the process of boosting the higher-frequency components of the message signal before FM modulation.

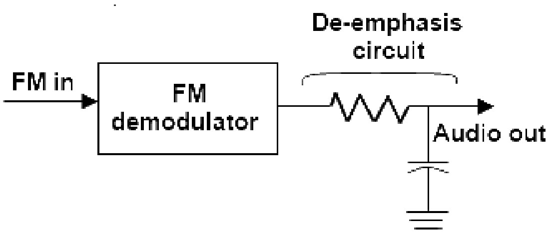


(a) Pre-emphasis Circuit

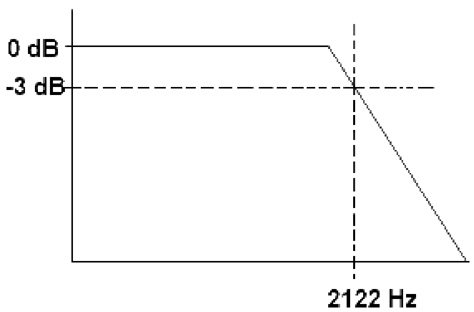


(b) Pre-emphasis Curve

De-emphasis is the process of attenuating the high-frequency components at the FM receiver after demodulation.



(c) De-emphasis circuit



(d) De-emphasis Curve